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Inventor(s)

Gao; Qiang et al.

Charging Device and Charging System

Abstract

A charging device includes a composite heat exchanger, a first electronic component, a second electronic component, and a housing. The composite heat exchanger includes a first heat exchange body and a second heat exchange body. The first heat exchange body has a heat conduction part and a heat dissipation part. The second heat exchange body has an evaporation end and a condensation end. The second heat exchange body has a cavity. A cooling medium is disposed in the cavity. The first electronic component is connected to the heat conduction part in a thermally conductive manner. The second electronic component is connected to the evaporation end in the thermally conductive manner. The housing is connected to the heat conduction part and the evaporation end, to form an airtight accommodating cavity. Both the first electronic component and the second electronic component are located in the accommodating cavity.

Inventors: Gao; Qiang (Dongguan, CN), Qiu; Yongqi (Shenzhen, CN)

Applicant: Huawei Digital Power Technologies Co., Ltd. (Shenzhen, CN)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This is a continuation of International Patent Application No. PCT/CN2023/101062 filed on Jun. 19, 2023, which claims priority to Chinese Patent Application No. 202211282017.1 filed on Oct. 19, 2022. The disclosures of the aforementioned applications are hereby incorporated by reference in their entireties.

TECHNICAL FIELD

[0002] This disclosure relates to the field of energy technologies, and in particular, to a charging device and a charging system.

BACKGROUND

[0003] With continuous development and popularization of new energy vehicles, increasingly more charging devices are also deployed. A charging device usually includes a plurality of different types of electrical components. Some electronic components generate a large amount of heat during operation. To ensure that the charging device is within a normal operating temperature range, heat dissipation usually needs to be performed on some electronic components in the charging device. Currently, air cooling and liquid cooling are usually used to dissipate heat for an electronic component. In the air-cooling manner, a fan is generally used to generate an airflow, and the airflow quickly passes through a surface of a heat generating device, to take away heat. However, in this manner, impurities such as dust are easily accumulated on the surface of the electronic component, and this affects reliability and a service life of the electronic component. The liquid cooling manner has high requirements on airtightness and reliability of a liquid cooling pipe, and has problems such as a complex structure, high construction difficulty, and high costs.

SUMMARY

[0004] This disclosure provides a charging device that features a simple structure and cost-effectiveness and can implement a high ingress protection rating, and a charging system.

[0005] According to a first aspect, this disclosure provides a charging device. The charging device may include a composite heat exchanger, a first electronic component, a second electronic component, and a housing. The composite heat exchanger includes a first heat exchange body and a second heat exchange body that are fastened. The first heat exchange body has a heat conduction part and a heat dissipation part. The first electronic component is connected to the heat conduction part in a thermally conductive manner. Therefore, heat generated by the first electronic component can be transferred to the heat dissipation part via the heat conduction part for heat dissipation, so as to cool the first electronic component. In addition, the second heat exchange body has an evaporation end and a condensation end. The second heat exchange body has a cavity. A cooling medium is disposed in the cavity. The cooling medium may circulate between the evaporation end and the condensation end. The second electronic component is connected to the evaporation end in the thermally conductive manner. After heat generated by the second electronic component is transferred to the evaporation end, the cooling medium located near the evaporation end in the cavity is evaporated due to the heat, and is condensed due to heat release at the condensation end. In other words, the second heat exchange body implements a heat dissipation function of the

second heat exchange body mainly based on the circulation of the cooling medium between the evaporation end and the condensation end. The housing is connected to the heat conduction part and the evaporation end, to form an airtight accommodating cavity. Both the first electronic component and the second electronic component are located in the accommodating cavity, so that airtightness of the first electronic component and the second electronic component can be ensured.

[0006] In an example, the heat conduction part is connected to the evaporation end in the thermally conductive manner, to improve heat dissipation efficiency of the composite heat exchanger. For example, in some cases, the heat of the first electronic component may be dissipated via the first heat exchange body and the second heat exchange body. In addition, the heat of the second electronic component may be dissipated via the second heat exchange body and the first heat exchange body. Therefore, a heat dissipation capability of the composite heat exchanger can be effectively used, improving the heat dissipation efficiency of the composite heat exchanger.

[0007] During specific disposition, the heat conduction part has a first contact surface, and the evaporation end has a second contact surface. The first electronic component is connected to the first contact surface in the thermally conductive manner. The second electronic component is connected to the second contact surface in the thermally conductive manner.

[0008] The first contact surface may be a plane or a curved surface. For example, the first contact surface may have a protrusion or a groove. The first electronic component may be connected to the groove or the protrusion in the thermally conductive manner, to appropriately adapt to the first electronic component of different sizes.

[0009] Certainly, during specific disposition, the second contact surface may also be a plane or a curved surface. Alternatively, the second contact surface may have a protrusion or a groove.

[0010] The first contact surface and the second contact surface may be disposed adjacent to each other, and the first contact surface and the second contact surface face a same direction. In this way, the first contact surface and the second contact surface can form a roughly flat surface.

[0011] Alternatively, in some examples, the first contact surface and the second contact surface may be disposed opposite to each other or disposed at an included angle.

[0012] In addition, in some examples, the heat dissipation part of the first heat exchange body may have a first recess part. The condensation end of the second heat exchange body is located in the first recess part, to reduce a volume of the entire composite heat exchanger, and facilitate miniaturization of the composite heat exchanger.

[0013] In some examples, the condensation end of the second heat exchange body may have a second recess part. The heat dissipation part of the first heat exchange body is located in the second recess part.

[0014] Alternatively, it may be understood that, during disposition, the first heat exchange body and the second heat exchange body may be disposed in a parallel location layout, or may be disposed in a stacked manner.

[0015] In addition, the heat dissipation part has a first airflow channel, and the condensation end has a second airflow channel. The charging device may further include a fan. The fan may enable air to flow through the first airflow channel and the second airflow channel quickly, to improve the heat exchange efficiency of the composite heat exchanger.

[0016] During specific disposition, the first airflow channel and the second airflow channel are connected, or the first airflow channel and the second airflow channel may be disposed in parallel.

[0017] When the first airflow channel is connected to the second airflow channel, the first airflow channel may be located downstream of the second airflow channel. In other words, cooling airflow may first pass through the second airflow channel and then pass through the first airflow channel, to preferentially ensure heat dissipation effect of the second electronic component.

[0018] In addition, in some examples, the first electronic component and the second electronic component may alternatively be connected to the housing in the thermally conductive manner. In other words, the heat generated by the first electronic component and the second electronic

component may alternatively be dissipated via the housing.

[0019] According to a second aspect, this disclosure further provides a charging system. The charging system may include the foregoing charging device and a charging terminal. The charging device may have an output port. The charging terminal may be connected to the output port. Electric energy processed by the charging device may be provided to the charging terminal, to supplement the electric energy to a powered device (like a vehicle).

[0020] During specific disposition, the charging device may be disposed on the ground, or may be disposed at a location like on a wall. A specific disposition location and form of the charging device are not limited in this disclosure.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0021] FIG. 1 is a diagram of an application scenario of a charging device according to an embodiment of this disclosure.

[0022] FIG. 2 is a diagram of a side structure of a charging device according to an embodiment of this disclosure.

[0023] FIG. 3 is a diagram of another side structure of a charging device according to an embodiment of this disclosure.

[0024] FIG. 4 is a top view of FIG. 3.

[0025] FIG. 5 is a diagram of still another side structure of a charging device according to an embodiment of this disclosure.

[0026] FIG. 6 is a top view of FIG. 5.

[0027] FIG. 7 is a diagram of yet another side structure of a charging device according to an embodiment of this disclosure.

[0028] FIG. 8 is a diagram of a structure of a location layout of a heat dissipation part and a condensation end according to an embodiment of this disclosure.

[0029] FIG. 9 is a diagram of an exploded structure of a composite heat exchanger according to an embodiment of this disclosure.

[0030] FIG. 10 is a diagram of a structure of a charging system according to an embodiment of this disclosure.

DESCRIPTION OF EMBODIMENTS

[0031] To make objectives, technical solutions, and advantages of this disclosure clearer, the following further describes this disclosure in detail with reference to accompanying drawings.

[0032] To facilitate understanding of a charging device provided in embodiments of this disclosure, the following first describes an application scenario of the charging device.

[0033] As shown in FIG. 1, in an example provided in this disclosure, a charging device **10** may be used in a scenario in which a vehicle **20** is charged. The charging device **10** may be a charging pile, and the charging device **10** may be disposed near a parking space. During actual application, an input end of the charging device **10** may be connected to a power grid **30**, and an output end of the charging device **10** may be connected to a charging connector **50** through a cable **40**, so as to charge the vehicle **20**. Electric energy in the power grid **30** may be industrial electric energy (for example, an alternating current voltage is 380 volts (V)), or may be residential electric energy (for example, an alternating current voltage is 22 V), or the like.

[0034] During actual application, the charging device **10** may have functions such as overload protection and power conversion. The charging device **10** usually includes electronic components such as a transistor, a transformer, an inductor, a capacitor, or a relay. Some electronic components generate heat during normal operation. To ensure that the charging device **10** is within a normal operating temperature range, heat dissipation usually needs to be performed on the electronic

components.

[0035] For a current charging device, heat dissipation is generally performed on the charging device in two different manners: air cooling and liquid cooling.

[0036] For example, in a charging device using the air-cooling manner, a fan is generally used to generate an airflow, and the airflow quickly passes through a surface of an electronic component, to take away heat. However, in this manner, impurities such as dust are easily accumulated on the surface of the electronic component, and this affects reliability and a service life of the electronic component.

[0037] In a charging device using the liquid cooling manner, this manner has high requirements on airtightness and reliability of a liquid cooling pipe, and has problems such as a complex structure, high construction difficulty, and high costs.

[0038] In view of this, embodiments of this disclosure provide a charging device **10** that features a simple structure and cost-effectiveness and can implement a high ingress protection rating, and a charging system in which the charging device **10** is used.

[0039] To make objectives, technical solutions, and advantages of this disclosure clearer, the following further describes this disclosure in detail with reference to the accompanying drawings and specific embodiments.

[0040] As shown in FIG. 2, in an example provided in this disclosure, the charging device **10** may include a composite heat exchanger **11**, a first electronic component **12**, a second electronic component **13**, and a housing **14**. The composite heat exchanger **11** includes a first heat exchange body **111** and a second heat exchange body **112** that are fastened. The first heat exchange body **111** has a heat conduction part **1111** and a heat dissipation part **1112**. The first electronic component **12** is connected to the heat conduction part **1111** in a thermally conductive manner. Therefore, heat generated by the first electronic component **12** can be transferred to the heat dissipation part **1112** via the heat conduction part **1111** for heat dissipation, so as to cool the first electronic component **12**. In addition, the second heat exchange body **112** has an evaporation end **1121** and a condensation end **1122**. The second heat exchange body **112** has a cavity **1123**. A cooling medium (not shown in FIG. 2) is disposed in the cavity **1123**. The cooling medium may circulate between the evaporation end **1121** and the condensation end **1122**. The second electronic component **13** is connected to the evaporation end **1121** in the thermally conductive manner. After heat generated by the second electronic component **13** is transferred to the evaporation end **1121**, the cooling medium located near the evaporation end **1121** in the cavity **1123** is evaporated due to the heat, and is condensed due to heat release at the condensation end **1122**. In other words, the second heat exchange body **112** implements a heat dissipation function of the second heat exchange body **112** mainly based on the circulation of the cooling medium between the evaporation end **1121** and the condensation end **1122**. The housing **14** is connected to the heat conduction part **1111** and the evaporation end **1121**, to form an airtight accommodating cavity **110**. Both the first electronic component **12** and the second electronic component **13** are located in the accommodating cavity **110**, so that airtightness of the first electronic component **12** and the second electronic component **13** can be ensured. In addition, during specific disposition, the first electronic component **12** and the second electronic component **13** may alternatively be connected to the housing **14** in the thermally conductive manner. In other words, the heat generated by the first electronic component **12** and the second electronic component **13** may alternatively be dissipated via the housing **14**.

[0041] In the example provided in this disclosure, the composite heat exchanger **11** includes two different types of heat exchange bodies, so that different heat dissipation characteristics and heat dissipation effect can be implemented. The first heat exchange body **111** dissipates heat mainly based on heat transfer of a material of the first heat exchange body **111**. The first heat exchange body **111** may be made of a material with good heat conductivity, like steel, copper, or aluminum. The first heat exchange body **111** has advantages of easy deployment, simple structure, cost-effectiveness, and high reliability. The second heat exchange body **112** dissipates heat mainly based

on air-liquid conversion of the cooling medium and the circulation of the cooling medium between the evaporation end **1121** and the condensation end **1122**. The second heat exchange body **112** has an advantage of high heat dissipation efficiency. Therefore, the second heat exchange body **112** may be configured to dissipate heat for an electronic component that generates a large amount of heat.

[0042] For example, in the charging device **10** provided in this disclosure, a heat flux density of the first electronic component **12** is less than a heat flux density of the second electronic component **13**. The heat flux density refers to heat energy passing through a unit area in unit time. In other words, in same time and per unit area, the heat generated by the first electronic component **12** is less than the heat generated by the second electronic component **13**. Therefore, during actual application, the heat conduction part **1111** of the first heat exchange body **111** may be connected to the first electronic component **12** in the thermally conductive manner, to dissipate heat for the first electronic component **12**, and reduce application costs of the composite heat exchanger **11**. The evaporation end **1121** of the second heat exchange body **112** may be connected to the second electronic component **13** in the thermally conductive manner, to dissipate heat for the second electronic component **13**, and ensure that the second electronic component **13** is within a normal operating temperature range.

[0043] In summary, in the charging device **10** provided in this disclosure, the composite heat exchanger **11** is used, so that deployment difficulty and application costs of a heat dissipation structure can be effectively reduced. In addition, high heat dissipation efficiency can be further achieved, so that a heat dissipation requirement of an electronic component that generates a large amount of heat can be met.

[0044] In addition, after the composite heat exchanger **11** is used, the first electronic component **12** and the second electronic component **13** may be disposed in the airtight accommodating cavity **110**. This can well protect the first electronic component **12** and the second electronic component **13**, and prevent the first electronic component **12** and the second electronic component **13** from being affected by impurities such as dust and water vapor in the air.

[0045] It should be noted that the foregoing first electronic component **12** and second electronic component **13** are for ease of differentiation and description. Types of the first electronic component **12** and the second electronic component **13** may be the same or may be different. This is not limited in this disclosure. In addition, during actual application, the charging device **10** may include a plurality of different types of electronic components, to implement a charging function of the charging device **10**.

[0046] For example, as shown in FIG. 3, a charging device **10** may include a first circuit board **113** and a plurality of first electronic components disposed on the first circuit board **113**. The plurality of first electronic components may be an inductor **12a**, an inductor **12b**, a capacitor **12c**, a capacitor **12d**, a transformer **12e**, and a transformer **12f**. In addition, the charging device **10** further includes a second circuit board **114** and a second electronic component **13** disposed on the second circuit board **114**. The second electronic component **13** may be a power component like an insulated-gate bipolar transistor (IGBT). A specific type of the second electronic component **13** is not limited in this disclosure.

[0047] The first circuit board **113** may be a printed circuit board (PCB), or may be a flexible printed circuit (FPC). In addition, types of the second circuit board **114** and the first circuit board **113** may be the same or may be different. Specific types of the first circuit board **113** and the second circuit board **114** are not limited in this disclosure. During actual application, a conductive connection may be implemented between the first circuit board **113** and the second circuit board **114** through a copper bar **115**, a cable, or the like.

[0048] In the example provided in this disclosure, the first circuit board **113** and the second circuit board **114** that are independent of each other are used, and all the first electronic components for which heat dissipation is performed via the first heat exchange body **111** are disposed on the first

circuit board **113**. This helps implement adaptation between the first heat exchange body **111** and the plurality of first electronic components, and facilitates disposition and positioning.

Correspondingly, the second electronic component **13** for which heat dissipation is performed via the second heat exchange body **112** is disposed on the second circuit board **114**. This helps implement adaptation between the second heat exchange body **112** and the second electronic component **13**, and facilitates disposition and positioning.

[0049] Certainly, in another example, the first circuit board **113** and the second circuit board **114** may alternatively be of an integrated structure, and details are not described herein. In addition, the charging device **10** may alternatively include two or more second electronic components **13**. A quantity of disposed first electronic components and a quantity of disposed second electronic components **13** are not limited in this disclosure.

[0050] During specific disposition, structure types of the first heat exchange body **111** and the second heat exchange body **112** are diversified.

[0051] For example, as shown in FIG. **3** and FIG. **4**, in an example provided in this disclosure, the first heat exchange body **111** is a serrate teeth heat exchanger. The first heat exchange body **111** includes a plate-shaped heat conduction part **1111** and a plurality of fins located on one side of the heat conduction part **1111**. The plurality of fins may be understood as a heat dissipation part **1112** of the first heat exchange body **111**. A surface that is of the heat conduction part **1111** and that is away from the fins may be used as a first contact surface **11110**, and the first electronic component (like the inductor **12a**) may be connected to the first contact surface **11110** in a thermally conductive manner. During specific disposition, the inductor **12a** may be connected to the first contact surface **11110** in the thermally conductive manner via a thermally conductive material (like thermally conductive silica gel), to ensure thermal conductivity efficiency between the inductor **12a** and the first contact surface **11110**.

[0052] The first heat exchange body **111** may be of an integrated structure or a split structure.

[0053] For example, the heat conduction part **1111** and the heat dissipation part **1112** may be integrally formed based on a process like die casting or numerical control machine tool processing. Alternatively, the heat conduction part **1111** and the heat dissipation part **1112** may be manufactured and molded separately, and then the heat conduction part **1111** and the heat dissipation part **1112** are connected based on a process like welding.

[0054] In addition, a gap between adjacent fins may form a first airflow channel **11120** for air circulation. When air flows in the first airflow channel **11120**, heat of the fins may be taken away. During specific disposition, a gap between fins and a quantity of fins may be appropriately selected based on an actual situation. This is not limited in this disclosure.

[0055] In the example provided in this disclosure, the plurality of first electronic components are all disposed on a same board surface (for example, an upper board surface in FIG. **3**) of the first circuit board **113**. Therefore, it is convenient to implement a thermally conductive connection between the plurality of first electronic components **12** and the first contact surface **11110**.

[0056] In addition, during specific disposition, because height sizes of different types of first electronic components are different, the first contact surface **11110** may be appropriately disposed for different height sizes of the first electronic components.

[0057] For example, in the example provided in this disclosure, the first contact surface **11110** has a plurality of protrusions. The first electronic component may be connected to the heat conduction part **1111** in the thermally conductive manner via the protrusions. A protrusion **11111** and a protrusion **11112** are used as an example. A height size of the inductor **12b** is greater than a height size of the capacitor **12c**. Therefore, during disposition, a height size of the protrusion **11111** may be less than a height size of the protrusion **11112**, to ensure that the inductor **12b** can be well connected to the protrusion **11111** in the thermally conductive manner, and that the capacitor **12c** can be well connected to the protrusion **11112** in the thermally conductive manner.

[0058] In the example provided in this disclosure, the first heat exchange body **111** has a simple

structure and is easy to be manufactured. Therefore, during application, a structural shape of the first heat exchange body **111** may be manufactured to be complex. For example, the protrusion **11111** and another part of the first heat exchange body **111** may be an integrated structure, and may be manufactured based on a die casting process, a cutting process, or the like. Certainly, in another example, the protrusion **11111** may alternatively be an independent mechanical part, and the protrusion **11111** may be fastened to the first contact surface through welding or the like.

[0059] In addition, in another example, the protrusion may alternatively be replaced with a groove structure, to adapt to different height sizes of different first electronic components. Details are not described herein again.

[0060] During specific application, the second heat exchange body **112** may be a loop thermosiphon (LTS). The second heat exchange body **112** is of a hollow thin-wall structure. One end (for example, a lower end in FIG. 3) of the second heat exchange body **112** may be used as an evaporation end **1121**. A surface of the evaporation end **1121** may be used as a second contact surface **11210** connected to the second electronic component **13** in the thermally conductive manner. In the example provided in this disclosure, a condensation end **1122** has a plurality of protrusion structures spaced from each other. A second airflow channel **11220** for air circulation may be formed between adjacent protrusion structures. When air flows in the second airflow channel **11220**, heat on a surface of the protrusion structure can be taken away. During specific disposition, a gap between protrusion structures and a quantity of protrusion structures may be appropriately selected based on an actual situation. This is not limited in this disclosure.

[0061] In addition, in the example provided in this disclosure, there is further a fin **11221** between the protrusion structures, and the fin **11221** may be configured to increase a heat exchange area between the condensation end **1122** and the outside, to help improve heat dissipation performance of the second heat exchange body **112**.

[0062] When the second heat exchange body **112** is disposed, a capillary structure (not shown in FIG. 3) may be further disposed in a cavity **1123** of the second heat exchange body **112**, to improve circulation efficiency of a cooling medium in the cavity **1123**, and help improve heat dissipation performance of the second heat exchange body **112**.

[0063] When the first heat exchange body **111** and the second heat exchange body **112** are disposed, a relative location between the first heat exchange body **111** and the second heat exchange body **112** may be various.

[0064] For example, as shown in FIG. 3 and FIG. 4, in an example provided in this disclosure, the first heat exchange body **111** and the second heat exchange body **112** may be disposed in parallel.

[0065] The first contact surface **11110** of the first heat exchange body **111** and the second contact surface **11210** of the second heat exchange body **112** may be disposed adjacent to each other, and directions of the first contact surface **11110** and the second contact surface **11210** may be approximately the same.

[0066] During actual application, a housing **14** may be connected to the first contact surface **11110** and the second contact surface **11210** in an airtight manner, to form an airtight accommodating cavity **110**. This improves convenience of connecting the housing **14** to the composite heat exchanger **11**.

[0067] In addition, a mixed-flow fan **116a** may be further disposed in the accommodating cavity **110**, so that a hot airflow in the accommodating cavity **110** can circulate and flow, and a problem of an excessively high local temperature is avoided.

[0068] In addition, as shown in FIG. 3, in an example provided in this disclosure, the first heat exchange body **111** may alternatively be connected to the second heat exchange body **112** in the thermally conductive manner.

[0069] The heat conduction part **1111** of the first heat exchange body **111** may be connected to the evaporation end **1121** of the second heat exchange body **112** in the thermally conductive manner, to improve overall heat dissipation efficiency of the composite heat exchanger **11**.

[0070] For example, when a temperature of the heat conduction part **1111** is higher than a temperature of the evaporation end **1121**, heat of the first heat exchange body **111** (or the first electronic component) may be dissipated via the second heat exchange body **112**. Alternatively, when a temperature of the evaporation end **1121** is higher than a temperature of the heat conduction part **1111**, heat of the second heat exchange body **112** (or the second electronic component **13**) may be dissipated via the first heat exchange body **111**. Alternatively, it may be understood that heat of the first electronic component **12** may be dissipated via the first heat exchange body **111** and the second heat exchange body **112**, so that a heat dissipation function of the composite heat exchanger **11** can be used to a large extent. Correspondingly, heat of the second electronic component **13** may be dissipated via the second heat exchange body **112** and the first heat exchange body **111**, to improve heat dissipation efficiency of the composite heat exchanger **11**.

[0071] During specific disposition, the heat conduction part **1111** may be welded to the evaporation end **1121**, to implement a fixed connection between the first heat exchange body **111** and the second heat exchange body **112**, and implement a thermally conductive connection between the heat conduction part **1111** and the evaporation end **1121**.

[0072] Certainly, in another example, the first heat exchange body **111** and the second heat exchange body **112** may alternatively be fastened via a connector like a screw. A connection manner between the first heat exchange body **111** and the second heat exchange body **112** is not limited in this disclosure.

[0073] It may be understood that, in another example, the thermally conductive connection may not be implemented between the first heat exchange body **111** and the second heat exchange body **112**, and details are not described herein again.

[0074] In addition, during specific disposition, the first airflow channel **11120** and the second airflow channel **11220** may be disposed in parallel or connected.

[0075] For example, as shown in FIG. 3 and FIG. 4, in an example provided in this disclosure, the first airflow channel **11120** and the second airflow channel **11220** are disposed in parallel. A fan **116b** and a fan **116d** may be disposed at two ends of the first airflow channel **11120** respectively, and a fan **116c** and a fan **116e** may be disposed at two ends of the second airflow channel **11220** respectively. The fan **116d** and the fan **116e** are of an air blowing type. In other words, the fan **116d** may be configured to blow air into the first airflow channel **11120**, and the fan **116e** is configured to blow air into the second airflow channel **11220**. The fan **116b** and the fan **116c** are of an air extraction type. In other words, the fan **116b** may be configured to extract air from the first airflow channel **11120**, and the fan **116c** may be configured to extract air from the second airflow channel **11220**.

[0076] In addition, a structure like a flow equalization plate (not shown in FIG. 3) may be disposed between the fans **116b** and **116d** and the first airflow channel **11120**, so that airflow generated by the fan **116b** and the fan **116d** can flow through each first airflow channel **11120** as evenly as possible.

[0077] Correspondingly, a structure like a flow equalization plate (not shown in FIG. 3) may be further disposed between the fans **116c** and **116e** and the second airflow channel **11220**, so that airflow generated by the fan **116c** and the fan **116e** can flow through each second airflow channel **11220** as evenly as possible.

[0078] During specific application, a quantity and locations of the fans may be appropriately set based on different requirements. For example, the fan **116d** and the fan **116e** may not be disposed. A quantity and locations of the disposed fans are not limited in this disclosure.

[0079] In addition, as shown in FIG. 5 and FIG. 6, in an example provided in this disclosure, the first airflow channel **11120** and the second airflow channel **11220** may be connected. During specific disposition, the first airflow channel **11120** may be located downstream of the second airflow channel **11220**. The cooling airflow may first pass through the second airflow channel **11220** to ensure heat exchange efficiency of the second heat exchange body **112**, and then pass

through the first airflow channel **11120** to perform heat dissipation on the first heat exchange body **111**.

[0080] It may be understood that, during actual application, the first heat exchange body **111** is generally configured to dissipate heat for an electronic component with a high heat flux density. Therefore, the first heat exchange body **111** needs to have good heat dissipation effect. Therefore, in the example provided in this disclosure, the second airflow channel **11220** is located upstream of the first airflow channel **11120**.

[0081] Certainly, in another example, the second airflow channel **11220** may alternatively be located downstream of the first airflow channel **11120**. During actual application, a relative location of the first airflow channel **11120** and the second airflow channel **11220** may be appropriately set based on an actual requirement, and details are not described herein again.

[0082] In addition, the first heat exchange body **111** and the second heat exchange body **112** may be disposed in parallel, or may be disposed in a stacked manner.

[0083] For example, as shown in FIG. 7 and FIG. 8, in another example provided in this disclosure, the first contact surface **11110** of the first heat exchange body **111** and the second contact surface **11210** of the second heat exchange body **112** are disposed opposite to each other.

[0084] During actual application, the housing **14** may be connected to the first contact surface **11110** and the second contact surface **11210** separately in an airtight manner, to form two airtight spaces. An accommodating cavity **110a** is formed between the housing **14** and the first contact surface **11110**, and an accommodating cavity **110b** is formed between the housing **14** and the second contact surface **11210**. The plurality of first electronic components, such as the inductor **12a**, the inductor **12b**, the capacitor **12c**, the capacitor **12d**, the transformer **12e**, and the transformer **12f**, are all disposed in the accommodating cavity **110a**, and are connected to the first contact surface **11110** in the thermally conductive manner. The plurality of second electronic components, such as the transistor **13a**, the transistor **13b**, and the transistor **13c**, are all disposed in the accommodating cavity **110b**, and are connected to the second contact surface **11210** in the thermally conductive manner. Different electronic components may be separately disposed in different accommodating cavities, and this helps improve flexibility of deploying the different electronic components.

[0085] In addition, a mixed-flow fan **116a** may be further disposed in the accommodating cavity **110a**, so that a hot airflow in the accommodating cavity **110a** can circulate and flow, and a problem of an excessively high local temperature is avoided. Certainly, in another example, a mixed-flow fan may alternatively be disposed in the accommodating cavity **110b**. This is not limited in this disclosure.

[0086] As shown in FIG. 9, in an example provided in this disclosure, the heat dissipation part **1112** has a first recess part **11121**, and the condensation end **1122** is located in the first recess part **11121**. In addition, the condensation end **1122** has a second recess part **11222**, and the heat dissipation part **1112** is located in the second recess part **11222**. Alternatively, it may be understood that the first heat exchange body **111** and the second heat exchange body **112** may be disposed in a cross manner, to reduce a volume of the entire composite heat exchanger **11**, and facilitate miniaturization of the composite heat exchanger **11**.

[0087] In the example provided in this disclosure, the first recess part **11121** is located in a middle region of the heat dissipation part **1112**. In other words, the first recess part **11121** is surrounded by the fin structures. The second recess part **11222** is located in a peripheral region of the condensation end **1122**. Therefore, when the condensation end **1122** is located in the first recess part **11121** and the heat dissipation part **1112** is located in the second recess part **11222**, the condensation end **1122** is surrounded by the fin structures of the heat dissipation part **1112**.

[0088] Certainly, in another example, disposition locations of the first recess part **11121** and the second recess part **11222** and a relative location between the heat dissipation part **1112** and the condensation end **1122** may alternatively be flexibly set based on different requirements, and

details are not described herein again.

[0089] In addition, as shown in FIG. 10, an embodiment of this disclosure further provides a charging system, including a charging terminal 60 and a charging device 10. The charging terminal 60 is connected to an output end of the charging device 10 through a cable. During specific application, the charging device 10 may perform processing like frequency conversion on an alternating current in a power grid, or may convert an alternating current into a direct current and then provide the direct current to the charging terminal 60. The charging terminal 60 may be connected to a powered device (for example, a vehicle), to supplement electric energy to the powered device. The charging terminal 60 may be in a form of a charging connector or the like. In addition, one charging device 10 may be connected to one charging terminal 60, or may be connected to a plurality of charging terminals 60. A quantity of charging terminals 60 is not limited in this disclosure.

[0090] During specific disposition, the charging device 10 may be disposed on the ground, or may be disposed at a location like on a wall. A specific disposition location and form of the charging device 10 are not limited in this disclosure.

[0091] In various embodiments of this disclosure, unless otherwise stated or there is a logic conflict, terms and/or descriptions in different embodiments are consistent and may be mutually referenced, and technical features in different embodiments may be combined based on an internal logical relationship thereof, to form a new embodiment.

[0092] “A plurality of” in this disclosure refers to two or more than two. “And/or” describes an association relationship between associated objects, and indicates that three relationships may exist. For example, A and/or B may indicate the following three cases: A exists alone, both A and B exist, and B exists alone, where A and B may be singular or plural.

[0093] It may be understood that various numbers in embodiments of this disclosure are merely used for differentiation for ease of description, and are not used to limit the scope of embodiments of this disclosure. The sequence numbers of the foregoing processes do not mean execution sequences, and the execution sequences of the processes should be determined based on functions and internal logic of the processes.

Claims

1. A charging device, comprising: a composite heat exchanger comprising: a first heat exchange body comprising: a heat conduction part; and a heat dissipation part; and a second heat exchange body fastened to the first heat exchange body and comprising: an evaporation end; a condensation end; a cavity; and a cooling medium disposed in the cavity and configured to circulate between the evaporation end and the condensation end; a first electronic component connected to the heat conduction part in a thermally conductive manner; a second electronic component connected to the evaporation end in the thermally conductive manner, wherein a first heat flux density of the first electronic component is less than a second heat flux density of the second electronic component; and a housing connected to the heat conduction part and the evaporation end to form an airtight accommodating cavity, wherein both the first electronic component and the second electronic component are located in the airtight accommodating cavity.
2. The charging device of claim 1, wherein the heat conduction part is connected to the evaporation end in the thermally conductive manner.
3. The charging device of claim 1, wherein the heat conduction part has a first contact surface, wherein the evaporation end has a second contact surface, wherein the first electronic component is connected to the first contact surface in the thermally conductive manner, and wherein the second electronic component is connected to the second contact surface in the thermally conductive manner.
4. The charging device of claim 3, wherein the first contact surface and the second contact surface

are disposed adjacent to each other, and wherein the first contact surface and the second contact surface face a same direction.

5. The charging device of claim 3, wherein the first contact surface and the second contact surface are disposed opposite to each other.

6. The charging device of claim 5, wherein the heat dissipation part has a first recess part, and wherein the condensation end is located in the first recess part.

7. The charging device of claim 5, wherein the condensation end has a second recess part, and wherein the heat dissipation part is located in the second recess part.

8. The charging device of claim 3, wherein the first contact surface has a groove or a protrusion, and wherein the first electronic component is connected to the groove or the protrusion in the thermally conductive manner.

9. The charging device of claim 1, wherein the heat dissipation part has a first airflow channel, and wherein the condensation end has a second airflow channel.

10. The charging device of claim 9, further comprising a fan, wherein the first airflow channel is connected to the second airflow channel, wherein the fan is configured to accelerate a flow speed of air in the first airflow channel and the second airflow channel, and wherein the first airflow channel is located downstream of the second airflow channel.

11. The charging device of claim 9, further comprising a first fan and a second fan, wherein the first airflow channel and the second airflow channel are arranged in parallel, wherein the first fan is configured to accelerate a first flow speed of air in the first airflow channel, and wherein the second fan is configured to accelerate a second flow speed of air in the second airflow channel.

12. The charging device of claim 1, wherein the first electronic component and the second electronic component are connected to the housing in the thermally conductive manner.

13. A charging system, comprising: a charging device comprising: an output port configured to be connected to a charging terminal; a composite heat exchanger comprising: a first heat exchange body fastened to the first heat exchange body and comprising: a heat conduction part; a heat dissipation part; and a second heat exchange body fastened to the first heat exchange body and comprising: an evaporation end; a condensation end; a cavity; and a cooling medium is disposed in the cavity and configured to circulate between the evaporation end and the condensation end; a first electronic component connected to the heat conduction part in a thermally conductive manner; a second electronic component connected to the evaporation end in the thermally conductive manner, wherein a first heat flux density of the first electronic component is less than a second heat flux density of the second electronic component; and a housing connected to the heat conduction part and the evaporation end to form an airtight accommodating cavity, wherein both the first electronic component and the second electronic component are located in the airtight accommodating cavity.

14. The charging system of claim 13, further comprising the charging terminal, wherein the heat conduction part is connected to the evaporation end in the thermally conductive manner.

15. The charging system of claim 13, wherein the heat conduction part has a first contact surface, wherein the evaporation end has a second contact surface, wherein the first electronic component is connected to the first contact surface in the thermally conductive manner, and wherein the second electronic component is connected to the second contact surface in the thermally conductive manner.

16. The charging system of claim 15, wherein the first contact surface and the second contact surface are disposed adjacent to each other, and wherein the first contact surface and the second contact surface face a same direction.

17. The charging system of claim 15, wherein the first contact surface and the second contact surface are disposed opposite to each other.

18. The charging system of claim 17, wherein the heat dissipation part has a first recess part, and wherein the condensation end is located in the first recess part.

19. The charging system of claim 17, wherein the condensation end has a second recess part, and

wherein the heat dissipation part is located in the second recess part.

20. The charging system of claim 15, wherein the first contact surface has a groove or a protrusion, and wherein the first electronic component is connected to the groove or the protrusion in the thermally conductive manner.
